

	HINDUSTAN PETROLEUM GREEN R&D CENTRE, KIADB INDUSTRIAL AREA, TARABAHALLI VILLAGE, HOSKOTE TALUK, BANGALORE -560067 Ph: 080-28078521/080-28078524
	MATERIAL SAFETY DATA SHEET- HP NanOKool

SECTION I - PRODUCT IDENTIFICATION	
Chemical Name	HP NanOKool
Chemical Application	Solar reflectance coating

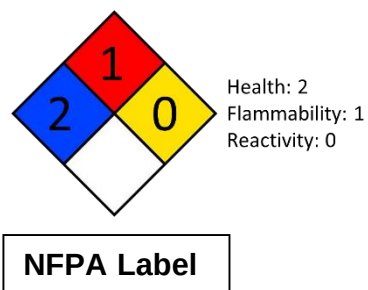
SECTION II - COMPOSITION/INFORMATION ON INGREDIENTS	
INGREDIENT(S)	Wt%
Acrylic Base	95%
Fillers	5%
	100%

The product has a shelf life of 2 year if it is stored in the original container between 18 to 25 °C. Also keep it away from humidity and excessive heat. Any concentration shown as a range is to protect the confidentiality or is due to batch variation.

SECTION III - HAZARDS IDENTIFICATION

The product has to be labeled due to the calculation procedure of international guidelines.

H222 : Extremely flammable aerosol.
 H229 : Pressurized container: May burst if heated.
 H319 : Causes serious eye irritation.
 H336 : May cause drowsiness or dizziness.



SECTION IV - FIRST AID MEASURES

General information:

Symptoms of poisoning may even occur after several hours; therefore, medical observation for at least 48 hours after the accident.

	HINDUSTAN PETROLEUM GREEN R&D CENTRE, KIADB INDUSTRIAL AREA, TARABAHALLI VILLAGE, HOSKOTE TALUK, BANGALORE -560067 Ph: 080-28078521/080-28078524 MATERIAL SAFETY DATA SHEET- HP NanOKool
---	--

After inhalation:

IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a POISON CENTER or doctor/physician if you feel unwell.

After skin contact:

Take off contaminated clothing and wash before reuse.

IF ON SKIN: Wash with plenty of soap and water. If skin irritation occurs: Get medical advice/attention.

After eye contact:

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/attention.

After swallowing:

If swallowed accidentally, do not induce vomiting and seek medical advice.

SECTION V - FIRE- FIGHTING MEASURES

Suitable extinguishing agents:

foam, carbon dioxide or dry agent.

Unsuitable Extinguishing Media:

None Known.

Protective equipment:

In case of fire and/or explosion do not breathe fumes. Protective respirator with independent air supply. According to size of fire Full protection, if necessary. Dispose of contaminated extinction water according to official regulations.

SECTION VI- ACCIDENTAL RELEASE MEASURES

Person-related safety precautions:

Use personal protective equipment. Ensure adequate ventilation. Remove all sources of ignition. Evacuate personnel to safe areas. Avoid contact with skin and eyes.

Measures for environmental protection:

Avoid disposing into drainage/sewer system or directly into the aquatic environment.

Measures for cleaning/collecting:

Absorb with liquid-binding material (sand, diatomite, acid binders, universal binders, sawdust).

	HINDUSTAN PETROLEUM GREEN R&D CENTRE, KIADB INDUSTRIAL AREA, TARABAHALLI VILLAGE, HOSKOTE TALUK, BANGALORE -560067 Ph: 080-28078521/080-28078524 MATERIAL SAFETY DATA SHEET- HP NanOKool
---	---

SECTION VII - HANDLING AND STORAGE

Handling:

Information for safe handling:

Ensure good ventilation/exhaustion at the workplace. Wash thoroughly after handling.

Information about protection against explosions and fires:

Keep ignition sources away.

Storage:

Store in cool place. Keep container tightly closed in a dry and well-ventilated place. Containers which are opened must be carefully resealed and kept upright to prevent leakage. **Storage**

Temperature: Store at 0 - 25 °C

SECTION VIII - EXPOSURE CONTROLS/PERSONAL PROTECTION

Engineering Controls	Good general ventilation should be sufficient to control worker exposure to airborne contaminants. Ensure eyewash stations and safety showers are close to the workstation location
PERSONAL PROTECTIVE EQUIPMENT	
Eyes and Face	Avoid contact with eyes. Wear chemical splash goggles or safety glasses with a side shield
Skin	Wear impermeable gloves and suitable protective clothing. Since this product is absorbed through the skin, care must be taken to prevent skin contact and contamination of clothing
Respiratory	Protective respiratory equipment may be used based on user's own risk assessment. Full Face Mask with particleType P2 filter.
Comments	Use a face shield, face mask, PVC gloves, and safety boots while handling

	HINDUSTAN PETROLEUM GREEN R&D CENTRE, KIADB INDUSTRIAL AREA, TARABAHALLI VILLAGE, HOSKOTE TALUK, BANGALORE -560067 Ph: 080-28078521/080-28078524 MATERIAL SAFETY DATA SHEET- HP NanOKool
---	---

SECTION IX - PHYSICAL AND CHEMICAL PROPERTIES

Appearance	Liquid
Color	White
Boiling Point/ range	193 °C
Flash Point	93.5 °C
Density (at 20°C)	1.206 g/cm ³
Solubility In Water	Miscible
Auto ignition temperature	459 °C

SECTION X - STABILITY AND REACTIVITY

Chemical Stability: Material is stable under normal conditions.

Materials to be avoided: Incompatible Materials

Dangerous reactions

No information

Dangerous products of decomposition:

Carbon dioxide and carbon monoxide

SECTION X - TOXICOLOGICAL INFORMATION

Toxicokinetics, metabolism and distribution:

Non-human toxicological data: Not available

Information on toxicological effects:

Acute toxicity:

Acrylic Polymer (CAS#9003-01-4):

LD50(Oral, Rat): ca. 357 mg/kg bw

LD50(Dermal, Rat): > 2000 mg/kg bw

LC50(Inhalation, Rat): > 5.1 mg/L 4 h

Skin corrosion/Irritation: Not classified.

Serious eye damage/irritation: Causes serious eye damage.

Respiratory or skin sensitization: Not classified

Germ cell mutagenicity: Not classified

Carcinogenicity: Not classified

Reproductive toxicity: Not classified

STOT- single exposure: May cause respiratory irritation.

STOT-repeated exposure: Not classified

Aspiration hazard: Not classified

SECTION XI - ECOLOGICAL INFORMATION

Ecological effects:

Do not allow product to reach groundwater, water course or sewage system.

SECTION XII - DISPOSAL CONSIDERATIONS

- Observe all central (Govt. of India), State, and local environmental regulations
- Must not be disposed of together with household garbage. Do not allow product to reach the sewage system

SECTION XIII - OTHER INFORMATION

The information and recommendations in literature and on labels are to the best of our knowledge reliable. However, nothing herein is to be construed as a warranty or representation. Users should make their own tests to determine the applicability of such information or the suitability of any products for their own particular purpose.

Disclaimer: Any concentration shown as a range is to project confidentially or is due to batch variation.

Further information

The information in the MSDS is based on data which is considered to be accurate. However, we makes no guarantees or warranty, either expressed or implied on the accuracy or completeness of this information. The conditions of handling, storage, use and disposal are beyond our control and may be beyond our knowledge. For this and other reasons, we do not assume responsibility and expressly disclaim liability for loss, damage or expense arising out of or in any way connected with the handling, storage, use or disposal of this product. This MSDS was prepared and is to be used for this product. If the product is used as a component in another product, this MSDS information may not be applicable.